Securing the Source: The Open-Weight Compliance and Maturity Index for the Global South

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1. Abstract

The governance of open-weight large language models (LLMs) is frequently paralyzed by a binary choice: strictly restrict access to guarantee safety, or release weights completely to democratize capabilities. For the Global South, restricting open-weight models exacerbates structural inequalities by enforcing reliance on costly, proprietary APIs. However, because built-in safety alignments ("guardrails") are easily stripped from open weights, releasing these models safely requires a new paradigm. Rather than treating safety as an all-or-nothing prerequisite, we propose the **Open-Weight Compliance and Maturity Index (OWMI)**. This framework shifts governance toward an incentive-driven model, scoring models based on the empirical evidence provided regarding upstream data curation, weight watermarking, and staged release testing. By replacing rigid gatekeeping with an evidence-based ranking system, we enable vulnerable nations to make informed, risk-adjusted procurement decisions while fostering a race to the top for open-weight safety transparency.

2. Introduction: The Need for Pragmatic Governance

For developers and institutions in the Global South, open-weight models are an infrastructural necessity to achieve technological sovereignty and avoid the recurring costs of closed APIs. However, current AI safety relies heavily on post-training alignment (like RLHF). Once weights are public, malicious actors can surgically remove these safety vectors through "abliteration" or localized fine-tuning, neutralizing standard safety protocols with minimal compute (Bowen et al., 2025).

While robust technical defenses exist—such as stripping dangerous knowledge from training data or embedding traceable watermarks—mandating perfect implementation prior to release stifles innovation, especially for decentralized developers lacking massive compute budgets. Governance must reflect this reality. We propose transitioning from binary release blocking to an empirical compliance index. This allows regional hubs (like BAISH) to rank models based on verifiable safety evidence, empowering local governments and hospitals to select open-weight models that match their risk tolerance.

3. Overcoming the Limitations of Current Frameworks

While the AI ecosystem has developed significant frameworks to measure transparency and manage risks, these were designed under paradigms that the OWMI seeks to transcend in order to address the technical and geopolitical realities of the Global South.

3.1. Beyond Transparency: OWMI vs. Model Openness Framework (MOF)

The *Model Openness Framework* (Linux Foundation) is an excellent standard for measuring transparency (how "open" a model is). However, it implicitly assumes that total openness is always safe. The OWMI evolves this concept by introducing **Resilience and Traceability**. While the MOF rewards openness for its own sake, the OWMI requires demonstrating that the open components include defenses against "abliteration" (the surgical removal of safety measures). OWMI does not just ask for transparency; it demands **auditable and safe openness**.

3.2. From Prohibition to Mitigation: OWMI vs. Preparedness Frameworks

Preparedness frameworks (such as those by OpenAI or Anthropic) operate under a binary *gatekeeping* model: if a model is dangerous, the release of weights is blocked and access is restricted to a controlled API. This centralized approach disproportionately harms the Global South, deepening technological dependence. OWMI starts from a pragmatic premise: **the release of weights is inevitable**. Therefore, instead of blocking the release, the OWMI mandates the application of *upstream* defenses (data curation) so that the model never acquires dangerous capabilities. OWMI converts a prohibitive barrier into an **empirical incentive index**.

3.3. From Documentary Audit to Empirical Validation: OWMI vs. Openness Indices

Current transparency indices (such as the *Artificial Analysis Openness Index*) are primarily based on documentary audits: they verify whether the documentation is correct or if the weights are available. The OWMI introduces **Empirical Auditing**. To achieve a high level in OWMI, it is not enough to publish a document; a watermark extraction key must be provided to independent auditors, or the absence of risks must be demonstrated using probe models. OWMI evaluates the effectiveness of the technical defense, not just the bureaucracy of publication.

4. The Three Pillars of Open-Weight Security

To build a functional maturity index, we must first define the technical pillars required to secure an open-weight model against adversarial misuse.

- **Pillar 1: Upstream Data Curation.** Because post-training alignment is easily bypassed, safety must start at the root. Curation ensures the training data mathematically lacks the latent space required to generate targeted harms (e.g., biological weapon synthesis). Even if safety guardrails are stripped, the model cannot retrieve knowledge it never learned.
- **Pillar 2: Model Provenance (Weight Watermarking).** If a model is weaponized, attributing the attack to a specific release is mathematically impossible without embedded signals (Cebrian et al., 2025). Deployers must embed training-free watermarks (like "EditMark") into the weights to trace adversarial fine-tuning back to its origin (Li et al., 2025).
- **Pillar 3: The Tiered Release Protocol.** Rejecting immediate, full-public releases in favor of a staged pipeline (structured auditing, limited beta under OpenRAIL licenses, and finally full release) to observe real-world fine-tuning behavior before irreversible proliferation.

5. The Core Framework: The Open-Weight Compliance and Maturity Index (OWMI)

The OWMI transforms the three pillars into a quantifiable ranking system. A model’s release tier and institutional trust rating are directly proportional to the empirical evidence the developers publish in an accompanying "Safety & Provenance Manifest".

Table 1: The OWMI Scoring Matrix

Compliance Pillar	Level 1: Foundational (Low Trust)	Level 2: Documented (Medium Trust)	Level 3: Empirical/Audited (High Trust)
Upstream Data Curation	Self-reported keyword filtering and basic deduplication	Publication of the semantic clustering and sanitization	Empirical validation via a secondary "probe" model

Compliance Pillar	Level 1: Foundational (Low Trust)	Level 2: Documented (Medium Trust)	Level 3: Empirical/Audited (High Trust)
	pipelines.	pipeline code used to isolate risk domains.	demonstrating the latent absence of targeted dangerous knowledge.
Weight Provenance	Simple metadata hashes and standard checksums (easily stripped).	Cryptographic weight watermarking applied, with methodology detailed in a whitepaper.	Verification key provided to independent auditors (e.g., BAISH); empirical proof that the watermark survives a 1-epoch fine-tuning.
Tiered Release & Testing	Immediate public upload to open repositories (e.g., Hugging Face).	Usage of OpenRAIL licenses and limited early access for institutional partners.	Independent red-teaming and abliteration-resistance report published by a third-party audit hub prior to public release.

6. Discussion: Incentives and "Sovereign Trust"

The primary strength of the OWMI is its incentive structure. It acknowledges that guaranteeing 100% perfect upstream data scrubbing on day one is nearly impossible for most developers. Instead, it creates a transparent "Sovereign Trust Ranking".

When governments, local healthcare providers, or enterprise systems in the Global South seek an open-weight model to fine-tune on sensitive regional data, they face massive risk. By adopting this index, regional AI safety hubs can evaluate incoming models. A model that achieves "Level 3" across the index—by providing clear, empirical evidence of its watermarking and data curation—becomes the default, trusted choice for institutional deployment. Conversely, models at "Level 1" are flagged as highly unstable for sensitive use cases.

6.1. Limitations and Future Directions

While the OWMI provides immediate pragmatic value, technical challenges remain. A known limitation of current weight watermarks is "lineage collapse"—provenance signals decay rapidly when open-weight models are repeatedly merged or heavily pruned by the community (arXiv:2605.24383). Future academic work must focus on *topology-resilient watermarks* (Luan et al., 2025) and optimizing extraction tools (Xue et al., 2025) to ensure Level 3 compliance can be maintained across multiple generations of model derivation.

7. Conclusion

For developing nations, open-weight AI is the primary mechanism for expanding sovereign AI capacity. By shifting the governance burden away from rigid gatekeeping and toward the **Open-Weight Compliance and Maturity Index**, we provide a scalable, incentive-driven roadmap. This framework maintains the democratizing power of open AI, rewards developer transparency, and equips vulnerable nations with the empirical tools needed to deploy AI safely.

References

- Bowen, D., Murphy, B., Cai, W., Khachaturov, D., Gleave, A., & Pelrine, K. (2025). Scaling Trends for Data Poisoning in LLMs. *arXiv*.
- Cebrian, M., Abeliuk, A., & Telle, J. A. (2025). Can adversarial attacks by large language models be attributed?. *arXiv*.
- Li, S., Chen, K., Jiang, J., Zhang, J., Yao, Q., Zeng, K., Zhang, W., & Yu, N. (2025). EditMark: Watermarking Large Language Models based on Model Editing. *arXiv*.
- Luan, Y., et al. (2025). Robust and Efficient Watermarking of Large Language Models Using Error Correction Codes. *Proceedings on Privacy Enhancing Technologies*.
- Xue, J., Zhao, Y., Al Ghanim, M., Gao, S., Sun, R., Lou, Q., & Zheng, M. (2025). PRO: Enabling Precise and Robust Text Watermark for Open-Source LLMs. *ICLR 2025 OpenReview*.
- Author's Note (arXiv:2605.24383). "A governance horizon for ethical-use constraints in open-weight AI models" (May 2026).